Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Cost-Effective Recombinant Epidermal Growth Factor Delivery with Enhanced Dermal Transport Property			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Protein Therapeutics	Industrial Biotechnology	Bio-Cosmetics/ materials
Concept of Proposed Technology	The proposed technology aims to develop a cost-effective recombinant Epidermal Growth Factor (EGF) using a microbial-based expression system optimized for scalable manufacturing. To enhance dermal applicability, the project incorporates a cell-penetrating peptide (CPP)–EGF fusion strategy, and assess the dermal transport efficiency of CPP-EGF in vitro and in vivo. The project integrates an RNA-assisted chaperone-based microbial expression platform for high-yield, soluble protein production with advanced delivery and tissue engineering approaches through collaboration with an overseas company. This integrated strategy supports formulation development, dermal delivery assessment, and pilot-scale feasibility toward future commercialization.			
Rationale for Joint R&D Project	This research project aims to address key technical challenges in the development of dermal permeable EGF, and validate dermal delivery feasibility and biological skin responses. The study leverages established expertise in biomaterials-based delivery systems and translational tissue engineering to systematically evaluate skin penetration mechanisms and regenerative responses. To enable practical translation of this research, collaboration with an industrial partner possessing a proprietary microbial expression platform for scalable and cost-effective recombinant EGF production is			

	essential. Through joint R&D, the project integrates complementary strengths, combining mechanistic evaluation of dermal delivery and skin regeneration with industrial-scale recombinant protein production and process development, thereby enhancing technical feasibility and global applicability.
Key Words	• Recombinant Epidermal Growth Factor (EGF) • Microbial-Based Protein Expression Platform • CPP-Modified EGF for Cosmeceutical Applications • Dermal Delivery and Skin Penetration Evaluation • Nanoparticle-Based Delivery and Formulation Systems

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)

R&D Objectives

Overall Objective

To advance recombinant EGF cosmeceutical technology from proof-of-concept to pilot-scale prototype production (TRL 6) through the integration of scalable microbial-based expression, optimized fermentation and purification processes, and stable formulation development suitable for dermal application.

Specific Objectives

- Establish a scalable recombinant EGF production system based on a microbial expression platform to enable consistent supply of well-defined protein materials for delivery and formulation studies.
- Optimize fermentation and purification processes to ensure product quality, consistency, and batch-to-batch reproducibility, providing a reliable foundation for mechanistic and translational evaluation.
- Design and develop nanoparticle-based formulation prototypes to enhance transdermal delivery efficiency, with emphasis on protein stability, controlled presentation, and dermal permeability.
- Validate pilot-scale production performance and process consistency in alignment with TRL 6 requirements, while assessing dermal delivery efficiency and potential regenerative skin responses using in vitro and in vivo models.

R&D Target Specifications

- **Protein:** Recombinant human Epidermal Growth Factor (EGF), produced as a well-defined protein material suitable for delivery and biomaterial-based formulation studies.
- **Molecular Weight:** Approximately **6.2 kDa**, consistent with native EGF and appropriate for controlled presentation in nanofiber and biomaterial systems.
- **Purity: ≥ 95%** (by validated analytical methods), enabling reliable mechanistic evaluation of delivery behavior and biological responses.
- **Biological Activity: ≥ 80%** relative to reference EGF, assessed by **in vitro cell proliferation assays**, ensuring functional relevance in skin and tissue-engineered models.
- **Structural Integrity:** Correct formation of **disulfide bonds**, confirmed by structural analysis, to support stability and bioactivity during formulation and delivery.
- **Endotoxin Level: ≤ 0.5 EU/mg**, suitable for in vitro translational studies involving skin-relevant cell and tissue models.
- **Stability: ≥ 12 months** under intended storage conditions, supporting reproducible formulation development and longitudinal evaluation of dermal delivery performance.

R&D Plan

Overall Research Approach

This project is structured to advance systematically from proof-of-concept studies to pilot-scale prototype production, with the goal of reaching TRL 6 by the end of the program.

The research integrates scalable microbial-based production of recombinant EGF, process optimization, and development of stable formulation prototypes, providing well-defined protein

	materials for delivery and translational evaluation in skin-relevant systems. The collaborative framework combines complementary expertise, linking recombinant protein production and pilot-scale process development with formulation design, dermal applicability assessment, and in vitro evaluation of biological responses. This integrated approach enables rigorous mechanistic investigation, reproducible performance assessment, and a clear translational pathway aligned with TRL 6 requirements. **Year 1 (TRL 2 → 3)**: Establishment of Defined Protein Materials and Initial Feasibility Studies - Establish recombinant EGF and CPP-EGF expression systems using a microbial platform to generate well-defined protein materials for delivery studies. - Evaluate protein solubility, yield, structural integrity, and biological activity to confirm suitability for formulation and mechanistic analysis. - Perform initial assessment of skin penetration feasibility using in vitro analytical and imaging models. - Assess compatibility of recombinant proteins with candidate formulation matrices, including biomaterial-based systems. **Year 2 (TRL 3 → 4–5)**: Process Optimization and Formulation-Driven Evaluation - Optimize fermentation and purification processes to ensure reproducible protein quality for delivery and translational studies. - Design and develop CPP-EGF formulation prototypes, including different nanoparticle-based systems. - Evaluate dermal delivery behavior, protein stability, and biological responses using in vitro and in vivo skin and tissue-engineered models. - Validate laboratory-scale prototypes to support progression toward pilot-scale translation. **Year 3 (TRL 5 → 6)**: Pilot-Scale Translation and Integrated Validation - Conduct pilot-scale production of recombinant EGF and CPP-EGF with controlled quality attributes. - Validate product quality, consistency, and dermal applicability in alignment with translational study requirements. - Perform final pilot-scale prototype evaluation to demonstrate technical readiness and alignment with TRL 6.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Systematic evaluation of whether and how CPP-EGF penetrates the skin barrier using nano-enabled imaging and analytical platforms • Design and assessment of nanoparticle-based delivery systems to improve dermal transport and protein stability • Assessment of regenerative skin responses using in vitro and in vivo tissue engineering models

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Expected R&R of Korean SME Partner	• Development and production of recombinant EGF and CPP-EGF constructs using the proprietary microbial expression platform • Optimization and scale-up of fermentation and purification processes • Pilot-scale production, quality control, and reproducibility assessment toward TRL 6

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** - Establishment of a pilot-scale recombinant EGF and CPP-EGF production process with validated performance, quality, and batch-to-batch consistency aligned with TRL 6 requirements. - Demonstration of an integrated production, purification, and formulation framework suitable for cosmeceutical applications, including assessment of stability and dermal applicability. **Industrial and Commercial Impact** - Enhanced competitiveness of recombinant EGF through cost-effective microbial-based pilot-scale manufacturing enabled by the proprietary expression platform. - Establishment of a technical foundation for future scale-up and downstream product development, supporting potential entry into domestic and global cosmeceutical markets. **Broader Impact** - Strengthening of bio-manufacturing and formulation development capabilities of Korean SMEs through pilot-scale process validation and international collaboration. - Promotion of international joint R&D in advanced bio-production, delivery evaluation, and translational tissue engineering technologies. - Demonstration of the transferability of the microbial production platform to other recombinant growth factors or bioactive proteins beyond EGF.

6. References

(1) Quality and cost assessment of a recombinant antibody fragment production process in *E. coli*. Kristell L. *et al*., New Biotechnology, 44; 31-40 (2018).

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(3) Protein solubility and folding enhancement by interaction with RNA. *Choi et al.*, PLoS One, 3(7): e2677 (2008).

(4) Playing catch-up with Escherichia coli: using yeast to increase success rates in recombinant protein production experiments. Roslyn MB. *et al.*, Front. Microbiol., 5: 85 (2014).